

Total metabolic tumor volume on ^{18}F -FDG PET/CT is a game-changer for patients with metastatic lung cancer treated with immunotherapy

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ABSTRACT

Purpose Because of atypical response imaging patterns in patients with metastatic non-small cell lung cancer (NSCLC) treated with immune checkpoint inhibitors (ICPIs), new biomarkers are needed for a better monitoring of treatment efficacy. The aim of this prospective study was to evaluate the prognostic value of volume-derived positron-emission tomography (PET) parameters on baseline and follow-up ^{18}F -fluoro-deoxy-glucose PET (^{18}F -FDG-PET) scans and compare it with the conventional PET Response Criteria in Solid Tumors (PERCIST).

Methods Patients with metastatic NSCLC were included in two different single-center prospective trials. ^{18}F -FDG-PET studies were performed before the start of immunotherapy (PET_{baseline}), after 6–8 weeks (PET_{interim1}) and after 12–16 weeks (PET_{interim2}) of treatment, using PERCIST criteria for tumor response assessment. Different metabolic parameters were evaluated: absolute values of maximum standardized uptake value (SUVmax) of the most intense lesion, total metabolic tumor volume (TMTV), total lesion glycolysis (TLG), but also their percentage changes between PET studies (ΔSUVmax , ΔTMTV and ΔTLG). The median follow-up of patients was 31 (7.3–31.8) months. Prognostic values and optimal thresholds of PET parameters were estimated by ROC (Receiver Operating Characteristic) curve analysis of 12-month overall survival (12M-OS) and 6-month progression-free survival (6M-PFS). Tumor progression needed to be confirmed by a multidisciplinary tumor board, considering atypical response patterns on imaging.

Results 110 patients were prospectively included. On PET_{baseline}, TMTV was predictive of 12M-OS [AUC (Area Under Curve) = 0.64; 95% CI: 0.61 to 0.66] whereas SUVmax and TLG were not. On PET_{interim1} and PET_{interim2}, all metabolic parameters were predictive for 12M-OS and 6M-PFS, the residual TMTV on PET_{interim1} (TMTV₁) being the strongest prognostic biomarker (AUC = 0.83 and 0.82; 95% CI: 0.74 to 0.91, for 12M-OS and 6M-PFS, respectively). Using the optimal threshold by ROC curve to classify patients into three TMTV₁ subgroups (0 cm³; 0–57 cm³; >57 cm³), TMTV₁ prognostic stratification was independent of PERCIST criteria on both PFS and OS, and significantly outperformed them. Subgroup analysis demonstrated that TMTV₁ remained a strong prognostic biomarker of 12M-OS for non-responding patients ($p=0.0003$) according to PERCIST criteria. In the specific

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Response evaluation in metastatic patients with non-small cell lung cancer (NSCLC) treated with immunotherapy is challenging because of atypical patterns that usual imaging response criteria (Response Evaluation Criteria in Solid Tumors V.1.1 for CT and positron-emission tomography (PET) Response Criteria in Solid Tumors (PERCIST) V.1.0 for ^{18}F -fluoro-deoxy-glucose PET (^{18}F -FDG PET/CT) fail to describe. New prognostic biomarkers were then necessary to better monitor response to immunotherapy.

WHAT THIS STUDY ADDS

⇒ The whole metabolic parameters are identified as prognostic factors on post-treatment ^{18}F -FDG PET-CT studies. PERCIST progressing patients with a low residual total tumorous volume after 6–8 weeks of immunotherapy have still a very favorable outcome (6-month progression-free survival = 73%; 24-month overall survival = 55%).

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ NSCLC metastatic patients with a low residual tumorous metabolic burden might have a benefit continuing immunotherapy after a first PET progression. PET metrics, especially metabolic tumor volume, might be included in predictive models.

group of patients with PERCIST progression on PET_{interim1}, low residual tumor volume (<57 cm³) was still associated with a very favorable patients' outcome (6M-PFS = 73%; 24M-OS = 55%).

Conclusion The absolute value of residual metabolic tumor volume, assessed 6–8 weeks after the start of ICPI, is an optimal and independent prognostic measure, exceeding and complementing conventional PERCIST criteria. Oncologists should consider it in patients with first tumor progression according to PERCIST criteria, as it helps identify patients who benefit from continued treatment.

Trial registration number 2018-A02116-49; NCT03584334.

INTRODUCTION

Non-small-cell lung cancers (NSCLC) account for 80% of lung cancers, which is the leading cause of cancer-related death worldwide.

Immune checkpoint inhibitors (ICPIs) targeting programmed death ligand-1 (PD-L1) and programmed cell death-1 (PD-1) receptors have been a breakthrough in the treatment of locally advanced or metastatic NSCLC and have led to significantly improved overall survival, with previously unseen rate of long-lasting response.^{1–3} However, less than half of patients will experience a clinical benefit to ICPIs.^{4,5} Identification of predictive biomarkers to early identify ICPI-responding patients is thus of utmost importance to optimize patient benefit, minimize the risk of toxicities and guide combination approaches. While the greatest focus has been on tumor-cell PD-L1 expression, it is an imperfect predictive biomarker as some patients with low/no PD-L1 levels respond to these drugs, and some patients with high PD-L1 expression do not.^{6–8} In the imaging field of research, previous studies have demonstrated baseline ¹⁸F-fluoro-deoxy-glucose positron-emission tomography (¹⁸F-FDG PET) metabolic parameters, such as tumor maximum standardized uptake value (SUVmax) and metabolic tumor volume (MTV), to be associated with patients' outcomes in the setting of ICPI, especially for NSCLC and melanoma.^{9–13}

New challenges have been also raised with the radiological monitoring of immunotherapy efficiency. Conventional imaging criteria such as Response Evaluation Criteria in Solid Tumors (RECIST) V.1.1, based on morphologic measurements on CT, or PET Response Criteria in Solid Tumors (PERCIST V.1.0), based on glucose metabolic changes on ¹⁸F-FDG-PET, rely on the paradigm that the cytotoxic effect of chemotherapy translates into measurable tumor shrinkage and metabolism decrease, respectively. Yet, the mechanisms of action of immunotherapy are different from conventional chemotherapy: the lifting of the inhibition of the lymphocyte system induces a tumor infiltration and proliferation of immune cells that can lead to a transient increase in tumor burden on CT, and metabolic activity on ¹⁸F-FDG-PET.¹⁴ Consequently, when using ICPI, new response patterns have been described such as a transient tumor increase in size and metabolism, or the appearance of new lesions, followed by a delayed response or stability. This challenging and specific immune-related response pattern is termed “pseudo-progression” (PsPD) and may lead to misinterpretation of clinical images. Previous studies have demonstrated that, in patients with NSCLC with a progressive disease under ICPI on the first PET-CT study, a subsequent PET could identify more than half of them with an atypical evolutive pattern such as PsPD or dissociated response, both patterns being associated with a favorable clinical outcome.^{15,16} Therefore, PERCIST criteria underestimate the efficacy of ICPIs, and new PET imaging criteria are needed to better distinguish between patients who benefit from ICPI and those who do not.^{17,18}

The aim of this study was to prospectively investigate the prognostic value of ¹⁸F-FDG PET metabolic parameters during the first 3 months of ICPIs treatment in patients with metastatic NSCLC, focusing on metabolic parameters derived from tumor volume.

MATERIALS AND METHODS

Participants

In this study, we collected data from two prospective observational studies conducted within our center (Centre Antoine Lacassagne, Nice), investigating the value of standard ¹⁸F-FDG PET/CT to monitor tumorous response to ICPIs in patients with metastatic NSCLC (FDG ECMI, n°ID-RCB: 2018-A02116-49; FDGIMMUN). These studies were approved by the ethics committees (respectively: CPP Sud Ouest et Outre Mer III—ID 759 N° 18.07.20.75329; CPP Sud-Est III—ID N° 18.07.16.43544) and all patients gave signed and informed consent to participate. For both studies, the inclusion criteria were (1) pathologically proven stage III B or IV NSCLC (2) an indication to start ICPIs in monotherapy in the first or later line according to current recommendations, (3) Eastern Cooperative Oncology Group (ECOG) performance status 0–2 and (4) age of at least 18 years. The exclusion criteria were (1) clinical or biological contraindication for ICPIs, (2) histological subtype other than adenocarcinoma, squamous cell carcinoma or undifferentiated carcinoma (3) delay higher than 3 months between PET_{baseline} and introduction of treatment; (4) prior treatment with ICPIs in the metastatic setting; (5) evidence of evolutive concurrent cancer; (6) vulnerable patients as defined in Article L1121-5 to L1121-8 of the French Public Health Code; (7) refusal of written consent; (8) high glycemia at baseline ¹⁸F-FDG-PET/CT (>9 mmol/L)⁹; no measurable lesion by PERCIST V.1.0.

Only metastatic patients were selected. Participants received one of the three possible treatment regimens: either pembrolizumab administered intravenously at a standard dose of 2 mg/kg or 200 mg every 3 weeks or nivolumab at a standard dose of 240 mg every 2 weeks or atezolizumab at a standard dose of 1200 mg every 3 weeks, following the guidelines.

The following clinical and biological parameters were recorded at baseline: age at treatment initiation, gender, weight, height, ECOG score, tobacco use, immunotherapy start date and molecule, prior treatments (surgery, radiotherapy, and systemic chemotherapy), tumor, node, metastasis staging, histological characteristics including histological type and pretreatment PD-L1 tumor expression when available.

Imaging protocol

¹⁸F-FDG PET/CT was performed before the start of ICPIs (PET_{baseline}), after 6–8 weeks of treatment (PET_{interim1}) and after 12–16 weeks of treatment (PET_{interim2}). A PET/CT was then scheduled every 3 months after the three initial PET/CT studies. Two different PET/CT imaging

systems were used: a Biograph mCT PET/CT scanner from February 2017 to September 2019 and a Biograph Vision 600 PET/CT scanner from September 2019 to January 2021 (Siemens Healthcare, Erlangen, Germany). Both PET systems fulfilled the EARL accreditation specifications for ^{18}F -FDG PET/CT tumor imaging. Patients were instructed to fast for at least 6 hours before the intravenous injection of 3 MBq/kg (Biograph mCT PET/CT) or 2.5 MBq/kg (Biograph Vision 600 PET/CT) of ^{18}F -FDG. A low-dose attenuation CT acquisition (80 kV, 50 mA, 5 mm slice thickness) was performed 60±5 min after the administration of ^{18}F -FDG, followed by an inspiratory chest-restricted diagnostic CT (auto-kV, auto-mA, 1-mm-slice thickness). Lastly, a diagnostic CT acquisition was done from the vertex of the skull to mid-thigh (auto-kV, auto-mA, 1-mm-slice thickness) after a venous injection of iodinated contrast agent in the absence of allergy or renal impairment. The same imaging system and the same acquisition parameters (duration and delay from injection) were used for baseline and post-treatment studies. PET images were reconstructed using the ordered subsets expectation maximization (OSEM 3D) iterative algorithm (2 iterations, 21 subsets), with point spread function and time-of-flight correction (ultra-HD PET). PET/CT systems main characteristics can be seen in online supplemental table 1.

Brain MRI was performed at baseline but was optional during follow-up and was performed only if brain lesions were detected at initial staging or if neurological signs appeared over time.

Image analysis

^{18}F -FDG PET/CT tumor metabolic parameters were measured using LifeX software¹⁹ (<https://www.lifexsoft.org>). To assess tumor burden, the total MTV was determined using a 4.0 SUV absolute fixed threshold (all voxels with SUV above the threshold were assigned to the tumor) on manually contoured hypermetabolic lesions. A downward adjustment of the threshold was exceptionally made for the few cases with lesions that were significantly hypermetabolic but had voxels with SUV below 4.0, typically for small lesions at baseline or small residual lesions on post-therapeutic studies, with SUVmax being underestimated by partial volume effect. A nuclear medicine resident performed the segmentation, and a 16 years experienced nuclear medicine physician reviewed it. In case of disagreement, segmentation was corrected by consensus. Pathological hypermetabolic lesions were included on the basis of the location and intensity of the uptake focus, its pattern (diffuse or focal) and the underlying morphological changes on the associated CT scan. Combined with their experience, these imaging features helped nuclear physicians to distinguish local inflammation from active tumor lesions. Pathological hypermetabolic foci obviously deemed to be due to the therapy-related inflammation or immune activation (eg, symmetrical uptake in enlarged hilar/mediastinal lymph nodes, diffuse splenic uptake, thyroiditis, colitis...) were

excluded of the lesion analyses. The SUVmax was calculated for the most intense lesion. Total lesion glycolysis (TLG) was obtained by multiplying the MTV by the mean SUV for each hypermetabolic lesion.

PET_{baseline}

The following parameters were assessed:

- ▶ SUVmax₀: SUVmax of the most intense lesion.
- ▶ Total metabolic tumor volume (TMTV₀): the whole MTV.
- ▶ SUVmean₀ of each hypermetabolic lesion.
- ▶ TLG₀: calculated as the sum of SUVmean multiplied by the metabolic volume of each lesion.

PET_{interim} 1 (after 6–8 weeks of treatment)

The following parameters were assessed:

- ▶ SUVmax₁: SUVmax of the most intense lesion.
- ▶ TMTV₁: the whole MTV (ie, the residual MTV).
- ▶ SUVmean₁ of each hypermetabolic lesion.
- ▶ TLG₁: calculated as the sum of SUVmean multiplied by the metabolic volume of each lesion.
- ▶ ΔSUV_1 corresponding to the SUV percentage change of the SUV_{max} of the most intense lesion (not necessarily the same) between PET_{baseline} and PET_{interim} 1, regardless of the appearance of new lesions: $\Delta\text{SUV}_1 (\%) = 100 \times (\text{SUVmax}_1 - \text{SUVmax}_0) / \text{SUVmax}_0$
- ▶ ΔTMTV_1 corresponding to the MTV percentage change between PET_{baseline} and PET_{interim} 1: $\Delta\text{TMTV}_1 (\%) = 100 \times (\text{TMTV}_1 - \text{TMTV}_0) / \text{TMTV}_0$.
- ▶ ΔTLG_1 corresponding to the TLG percentage change between PET_{baseline} and PET_{interim} 1: $\Delta\text{TLG}_1 (\%) = 100 \times (\text{TLG}_1 - \text{TLG}_0) / \text{TLG}_0$.
- ▶ PERCIST criteria but also the iPERCIST (immune PET Response Criteria in Solid Tumor) criteria, adapted to the issue of immunotherapy and inspired by previous guidelines and studies were used for the interpretation of the PET scans (online supplemental table 2).
- ▶ RECIST V.1.1 criteria and iRECIST (immune Response Evaluation Criteria in Solid Tumor) criteria were evaluated in accordance with published guidelines, using whole-body diagnostic CT scan acquisition and chest-restricted inspiratory diagnostic CT, both performed at the time of the PET/CT study.^{20 21}

PET_{interim} 2 (after 12–16 weeks of treatment)

The following parameters were assessed, using the same definitions as on PET_{interim} 1: SUVmax₂, TMTV₂, SUVmean₂, TLG₂, ΔSUV_2 , ΔTMTV_2 , ΔTLG_2 , PERCIST, iPERCIST, RECIST and iRECIST criteria.

Follow-up and clinical endpoints

Patients were followed for at least 12 months with regular clinical evaluations and standard-of-care imaging (including follow-up brain MRI, ^{18}F -FDG PET and CT studies).

The delay between the initiation of the treatment and the decision to stop it, as well as the reason for this stop

(tumor progression, toxicity, therapeutic break, patient's refusal to continue the treatment) were recorded.

The primary endpoint was overall survival (OS) defined as the time from initial immunotherapy to death from any cause.

Progression-free survival (PFS) was a secondary endpoint of the study and was defined as the time from the initiation of ICPI to confirmed tumor progression or death. Tumor progression needed to be confirmed by a multidisciplinary tumor board, confronting the patient's clinical status, PET and CT image interpretation, and brain MRI results. Thus, treatment decisions rest with the entire healthcare team, as suggested by previous guidelines.²¹ Confirmed tumor progression necessarily implied the decision to definitively stop the treatment. For PFS, we did not take the time of first evidence of tumor progression on PET/CT due to PsPD or dissociated response patterns.

Statistical analysis

Statistical analysis was performed using R, V.3.6.3, by DC. Continuous variables are summarized as medians and IQRs and categorical variables are summarized as numbers of patients and percentages. For continuous variables, statistical significance was determined using Student's t-test or Wilcoxon's test. A p value of 0.05 or less was considered significant.

The prognostic values of the different PET parameters were estimated through ROC (Receiver Operating Characteristic) curve analysis. Optimal cut-off values for each parameter were obtained with these ROC curve analyses. Univariate analysis was also performed and all parameters

with a p value < 0.1 were then entered in a multivariate forward stepwise cox regression analysis.

Survival curves stratified by response at first and second evaluations were constructed by the Kaplan-Meier method and compared by use of the log-rank test. Patients were censored after 36 months of follow-up or at the date of the latest news.

AUC (Area Under Curve) values were compared using a DeLong or a Bootstrap test.

RESULTS

Patients' characteristics

Of the 135 eligible patients, 110 were finally included in this ancillary study, from October 2016 to October 2020. Others were excluded from the study because they had a non-metastatic stage (N=9), non-eligible histological type (N=3), they were lost to follow-up before 1 year (N=3), they had a concurrent evolutive cancer (N=1), immunotherapy was finally non-initiated after patient's inclusion (N=4), poor image quality (N=1), economical reason (N=1), delay between PET_{baseline} and introduction of ICPIs was higher than 3 months or delay between introduction of ICPIs and PET_{interim1} was higher than 10 weeks (N=3). 15 of the 110 included patients did not undergo the PET_{interim1} study because of early discontinuation of ICPI (figure 1). Patients' characteristics at baseline are shown in table 1.

All patients had a metastatic NSCLC. The patients were mainly men (64%), median patient age was 63 years (range: 39–91). The histological subtype was squamous cell carcinoma for 16% (18/110) and non-squamous cell

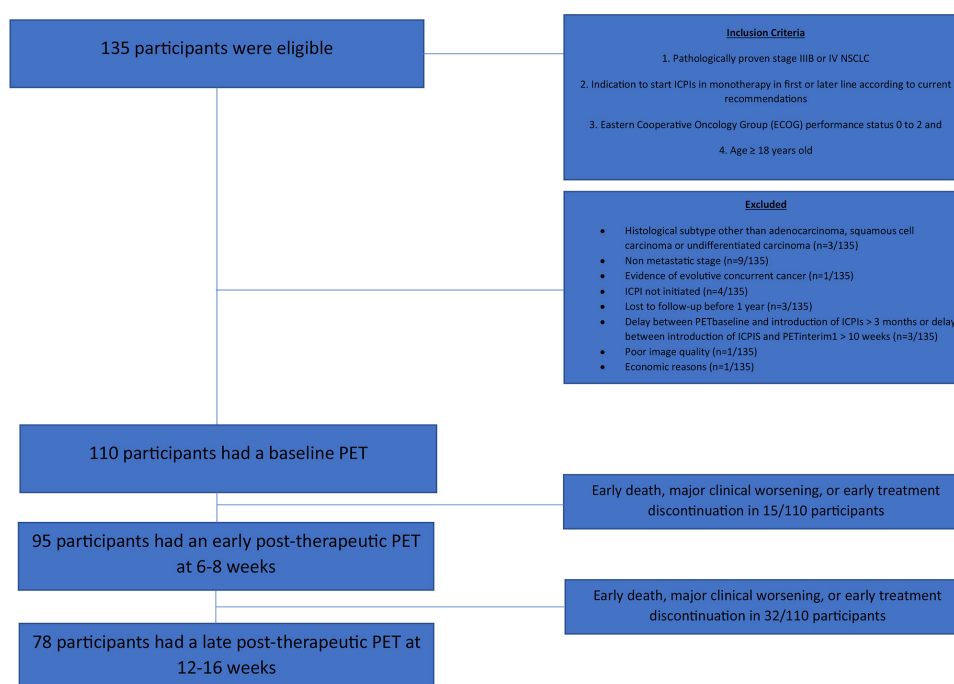


Figure 1 Flow-chart. ICPIs, immune checkpoint inhibitors; NSCLC, non-small-cell lung cancers; PET, positron-emission tomography.

Table 1 Patients' characteristics at baseline

	N	Per cent
Total patients	110	100
Age in years, mean (range)	63 (39–91)	
Sex		
Male	70	64
Female	40	36
Histology		
Squamous cell carcinoma	18	16
Non-squamous cell carcinoma	92	84
WHO performance status		
0	28	26
1	63	57
2	19	17
Current or former smoker		
Yes	88	80
No	22	20
Stage		
IV	110	100
Previous lines of chemotherapy		
None	33	30
1	39	35
2	24	22
≥3	14	13
Prior surgery	17	15
Prior radiotherapy	38	35
PD-L1 expression		
0	11	10
1–49	33	30
≥50	49	45
Unknown	17	15
Treatment		
Pembrolizumab	67	61
Nivolumab	39	35
Atezolizumab	4	4
PD-L1, programmed death ligand-1 .		

carcinoma for 84% (92/112) of patients. PD-L1 expression was known for 85% (95/110) of patients. 100% of patients were stage IV (110/110) at baseline. 30% were treated with ICPI as the first line (33/110) while 70% had undergone previous lines of chemotherapy (77/110) before receiving ICPIs. 15% had had prior surgery (17/110) and 35% had had prior radiotherapy (38/110), before restaging at the time of immunotherapy. 61% of patients were treated with pembrolizumab (67/110), 35% with nivolumab (39/110) and 4% with atezolizumab (4/110). PFS and OS curves according to ECOG performance status can be seen in online supplemental figure 1.

The median follow-up was 31 months (IQR 7.3–31.8 months). All patients (n=110) were evaluated with a baseline ^{18}F -FDG-PET/CT study. The median delay between baseline PET and the introduction of immunotherapy was 7 days (IQR=2–22 days; range=0–74 days). The median delay between the introduction of immunotherapy and PET_{interim} 1 was 50 days (IQR 48–52 days). The median delay between the introduction of immunotherapy and PET_{interim} 2 was 92 days (IQR 91–99).

Metabolic parameters and response evaluations on baseline and interim PET/CT studies

The PET_{baseline} study was available for all patients (n=110). Baseline PET/CT examinations were performed on the Biograph mCT PET/CT system for 79 patients and on the Biograph Vision 600 PET/CT system for 31 patients. All but six patients were evaluated on the same PET/CT systems for follow-up studies.

Median SUV_{max}₀, TMTV₀ and TLG₀ were 10.0 (6.8–14.3), 39.2 (9.1–108.1) and 196.7 (43.8–414.5), respectively.

Baseline SUV_{max} was not significantly different between the two PET/CT systems (mean: 11.2±6.6 vs 12.8±9.5, p=0.37).

The PET_{interim} 1 study was available for 95 out of 110 patients (86%). In the 15 remaining patients, it was waived due to early death (n=10) or major clinical worsening (n=5) that did not allow ICPI to be continued. Median SUV_{max}₁, TMTV₁ and TLG₁ were 9.2 (5.8–12.4), 19.9 (5.3–102) and 101.5 (23.7–551.2), respectively. Median ΔSUV₁, ΔTMTV₁ and ΔTLG₁ were −6.3% (−30.5–16.3), 1% (−79–84) and −3.3% (−82.9–99.5), respectively.

Tumor response on PET_{interim} 1 according to (i)PERCIST criteria:

- ▶ 8% of patients had a complete metabolic response (n=8/95).
- ▶ 19% of patients had a partial metabolic response (n=18/95).
- ▶ 10% of patients had a stable metabolic disease (n=9/95).
- ▶ 63% of patients had a progressive metabolic disease using PERCIST criteria (n=60/95), equivalently classified as unconfirmed progressive metabolic disease (uPMD) using iPERCIST criteria. ICPI was continued over this first progression for six more weeks, with the exception of patients whose treatment was prematurely discontinued due to early death (5/60) or major clinical worsening (10/60) that did not allow ICPI to be continued according to the multidisciplinary tumor board. For one patient, treatment was continued but PET_{interim} 2 was omitted.

Tumor response on PET/CT_{interim} 1 according to RECIST criteria:

- ▶ 1% of patients had a complete response (n=1/95).
- ▶ 13% of patients had a partial response (n=12/95).
- ▶ 44% of patients had a stable disease (n=42/95).

- 42% of patients had a progressive disease (n=40/95) equivalently classified as immune unconfirmed progressive disease using iRECIST criteria.

The PET_{Interim}2 study was available for 78 out of 110 patients (71%). PET was waived in the remaining 32 patients due to early death (n=13), major clinical worsening (n=14), toxicity (n=1) and PETinterim2 omission (n=4).

Median SUVmax₂, TMTV₂ and TLG₂ were 7.4 (4.8–10.5), 14.0 (4.5–74.1) and 106.3 (20.9–336.5), respectively.

Median Δ SUV₂, Δ TMTV₂ and Δ TLG₂ were –23% (–56–12.0), –27.0% (–93.3–60.3) and –44.7% (–95.8–55.7), respectively.

Tumor response on PET_{Interim}2 using PERCIST criteria:

- 12% of patients had a complete metabolic response (n=9/78).
- 24% of patients had a partial metabolic response (n=19/78).
- 10% of patients had a stable metabolic disease (n=8/78).
- 54% of patients had a progressive metabolic disease (n=42/78).

Tumor response on PET_{Interim}2 using iPERCIST criteria:

- 12% of patients had a complete metabolic response (n=9/78).
- 24% of patients had a partial metabolic response (n=19/78).

- 10% of patients had a stable metabolic disease (n=8/78).
- 39% of patients had a confirmed progressive metabolic disease (n=30/78).
- 15% of patients had an unconfirmed progressive metabolic disease (n=12/78).

Of the 60 patients with uPMD on the first PET_{interim}1, 25% (15/60) had a subsequent PERCIST response or stability on PETinterim2, confirming a posteriori PsPD. In this subgroup of 15 patients, 6 showed a decrease in TMTV on PET_{interim}1 (despite PERCIST PMD), whereas 1 had stable TMTV (+2%) and 8 had an increase in TMTV.

Predictive and prognostic value of ¹⁸F-FDG PET/CT quantitative metrics

The prognostic accuracy of the different metabolic parameters in ROC curves, to predict 6 months PFS and 12 months OS, are shown in [table 2](#) and [figure 2](#).

Using ROC curves analyses to predict 12months OS ([table 2](#), [figure 2](#)):

- At PET_{baseline}, only TMTV₀ (AUC=0.64; 95% CI: 0.53 to 0.75) could moderately predict 12months OS. In contrast, SUVmax₀ and TLG₀ could not.
- At PET_{interim}1, all the metabolic parameters had significant predictive power for 12months OS, TMTV₁ having the best performance (AUC=0.83; 95% CI: 0.74 to 0.91; cut-off value=57 cm³).

Table 2 Prognostic value of metabolic parameters

Parameter	6 months PFS			12 months OS		
	AUC	95% CI	Opt. cut-off value	AUC	95% CI	Opt. cut-off value
PET _{baseline}						
SUV ₀	0.51	0.401 to 0.618		0.55	0.435 to 0.656	
TMTV ₀ (cm ³)	0.61	0.504 to 0.719		0.64	0.529 to 0.748	53.1
TLG ₀	0.59	0.479 to 0.694		0.61	0.504 to 0.724	
PET _{interim} 1						
SUV ₁	0.71	0.603 to 0.809	8.1	0.69	0.587 to 0.799	8.3
TMTV ₁ (cm ³)	0.82	0.739 to 0.909	57.5	0.83	0.742 to 0.910	57.5
TLG ₁	0.81	0.720 to 0.898	288.6	0.81	0.722 to 0.898	256.3
Δ SUV ₁ (%)	0.76	0.659 to 0.855	–21.5	0.74	0.644 to 0.843	4.4
Δ TMTV ₁ (%)	0.83	0.749 to 0.917	17	0.78	0.688 to 0.873	–38
Δ TLG ₁ (%)	0.83	0.740 to 0.912	14.9	0.77	0.671 to 0.862	–43.5
PET _{interim} 2						
SUV ₂	0.75	0.646 to 0.861	7.2	0.72	0.598 to 0.838	6.6
TMTV ₂ (cm ³)	0.86	0.756 to 0.965	31.4	0.77	0.646 to 0.897	45.3
TLG ₂	0.85	0.739 to 0.952	153.2	0.75	0.617 to 0.879	110.6
Δ SUV ₂ (%)	0.73	0.616 to 0.848	–24.5	0.68	0.558 to 0.807	–46
Δ TMTV ₂ (%)	0.84	0.733 to 0.946	–31.7	0.68	0.550 to 0.808	–51
Δ TLG ₂ (%)	0.84	0.736 to 0.948	–43.4	0.66	0.532 to 0.793	–86.9

ROC curve analyses of PET parameters for prediction of OS and PFS.

AUC, Area Under Curve; OS, overall survival; PET, positron-emission tomography; PFS, progression-free survival; ROC, Receiver Operating Characteristic; SUVmax, maximum standardized uptake value; TLG, total lesion glycolysis; TMTV, total metabolic tumor volume.

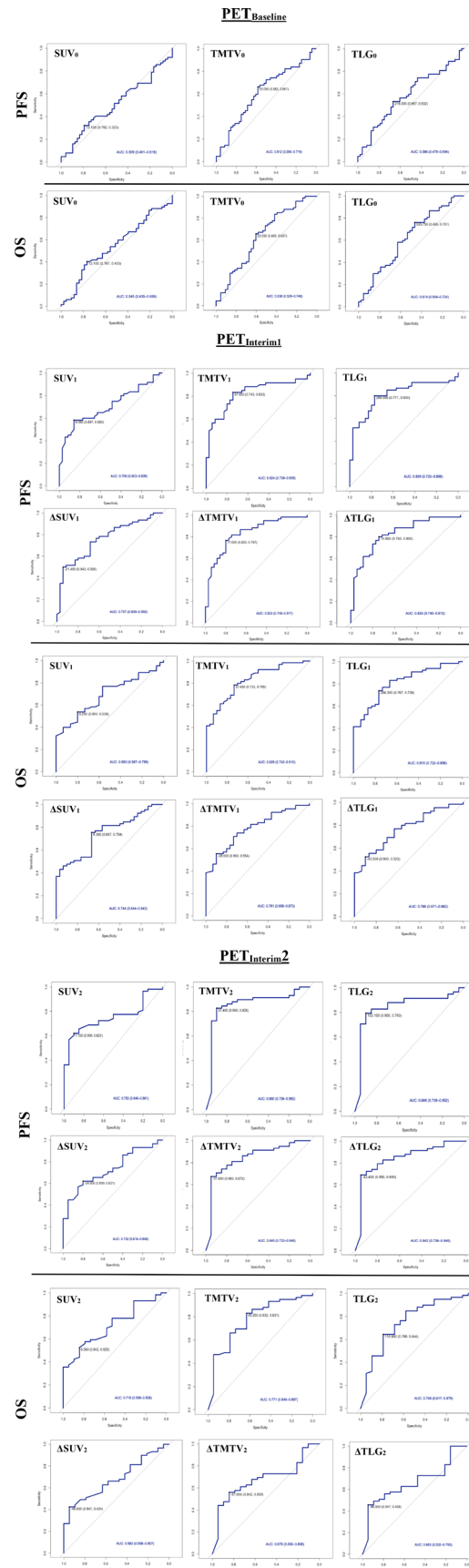


Figure 2 ROC curves figures of metabolic parameters at PET_{baseline}, PET_{interim1} and PET_{interim2}, to predict 6 months PFS and 12 months OS. OS, overall survival; AUC, Area Under Curve; PET, positron-emission tomography; PFS, progression-free survival; ROC, Receiver Operating Characteristic; SUV, standardized uptake value; TLG, total lesion glycolysis; TMTV, total metabolic tumor volume.

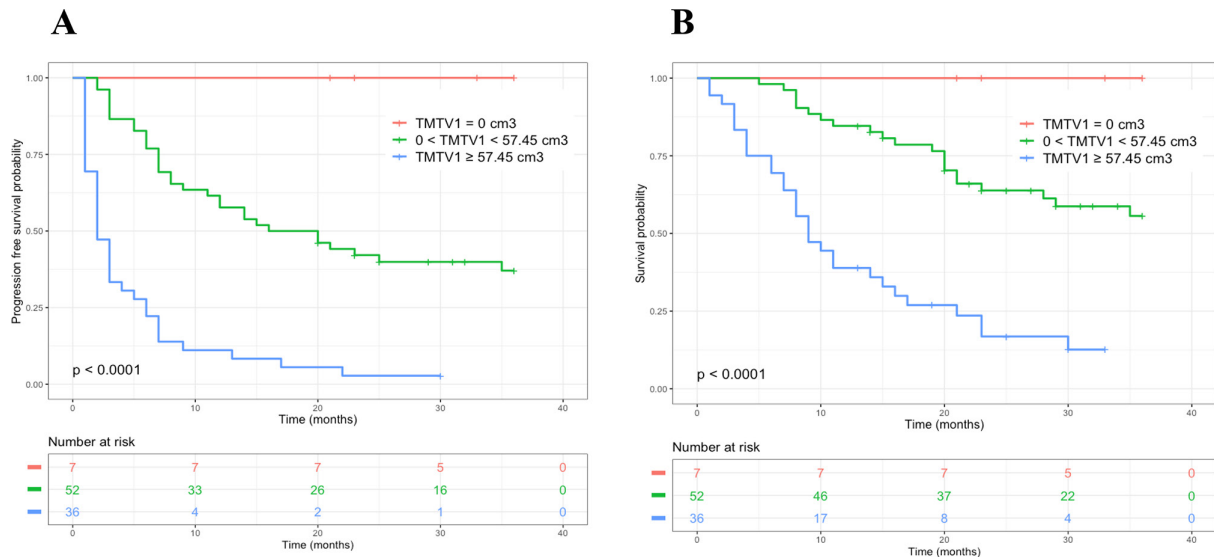


Figure 3 Progression-free survival (A) and overall survival (B) according to TMTV₁. MTV, metabolic tumor volume; TMTV, total metabolic tumor volume.

- At PET_{interim} 2, all the metabolic parameters had significant predictive power for 12months OS, TMTV₂ having the best performance (AUC=0.77; 95% CI: 0.65 to 0.90; cut-off value=45 cm³).

Using ROC Curves analyses to predict 6months PFS (table 2, figure 2):

- At PET_{baseline}, none of the metabolic parameters could predict 6months PFS.
- At PET_{interim} 1, all the metabolic parameters had significant predictive power for 6months PFS. Among them, TMTV₁ (AUC=0.82; 95% CI: 0.74 to 0.91), ΔTMTV₁ (AUC=0.83; 95% CI: 0.75 to 0.92) and ΔTLG₁ (AUC=0.83; 95% CI: 0.74 to 0.91) had the best performances.
- At PET_{interim} 2, all the metabolic parameters had significant predictive power for 6months PFS. Among them, TMTV₂ (AUC=0.86; 95% CI: 0.76 to 0.97), TLG₂ (AUC=0.85; 95% CI: 0.74 to 0.95), ΔTMTV₂ (AUC=0.84; 95% CI: 0.73 to 0.95) and ΔTLG₂ (AUC=0.84; 95% CI: 0.74 to 0.95) had the best performances.

Using the optimal threshold by ROC curves to classify patients into three TMTV₁ subgroups (0 cm³; 0–57 cm³; >57 cm³) (figure 3):

- The 6months PFS rates were 100%, 76.9% and 22.2% for patients with complete metabolic response, low residual TMTV₁ and high residual TMTV₁, respectively (p<0.001).
- The 6months OS rates were 100%, 98.1% and 69.4% for patients with complete metabolic response, low residual TMTV₁ and high residual TMTV₁, respectively.
- The 24months PFS rates were 100%, 39.9% and 2.8% for patients with complete metabolic response, low residual TMTV₁ and high residual TMTV₁, respectively.
- The 24months OS rates were 100%, 63.8% and 16.8% for patients with complete metabolic response, low residual TMTV₁ and high residual TMTV₁, respectively.

This classification, based on TMTV₁, provided a better outcome stratification than the conventional PERCIST criteria on survival curves (figures 3 and 4). For instance, 24months OS rates were 16.8% for patients with high TMTV₁ volume versus 33.7% for patients with a PERCIST progression on PET_{interim} 1 (p=0.029) and 6months PFS rates were 22% for patients with high TMTV₁ volume versus 43% for patients with a PERCIST progression on PET_{interim} 1 (p=0.019).

PFS and OS curves according to median values of SUVmax and TMTV at the different time points can be seen in online supplemental figure 2 with analogous findings.

To complement these results, the PFS and OS curves according to RECIST criteria are shown in online supplemental figure 3. At first evaluation, RECIST criteria enabled a very good prognostic stratification in the “partial response” group (6-month PFS (6M-PFS)=91.7%; 24-month OS (24M-OS)=90.9%) and the RECIST “progressive disease” group (6M-PFS=27.5%; 24M-OS=23.4%). However, 44.2% (42/95) of patients belonged to the “stable disease” response group, with an intermediate prognosis (6M-PFS=76.2%; 24M-OS=59.7%). In contrast, only 10% (10/98) of patients were classified in the “stable metabolic disease” group according to PERCIST criteria.

PFS and OS curves according to iRECIST and iPERCIST evaluation at PET_{interim} 1 can be seen in online supplemental figure 4.

Multivariate analyses showed that, among the evaluated clinico-biological and imaging characteristics of patients, TMTV₁ and PERCIST criteria were two independent prognostic parameters on 6M-PFS (p<0.001 and p=0.005, respectively) and TMTV₁ was an independent prognostic factor on 12M-OS (p=0.004) (online supplemental table 3).

Combining PERCIST criteria and TMTV₁ (figure 5):

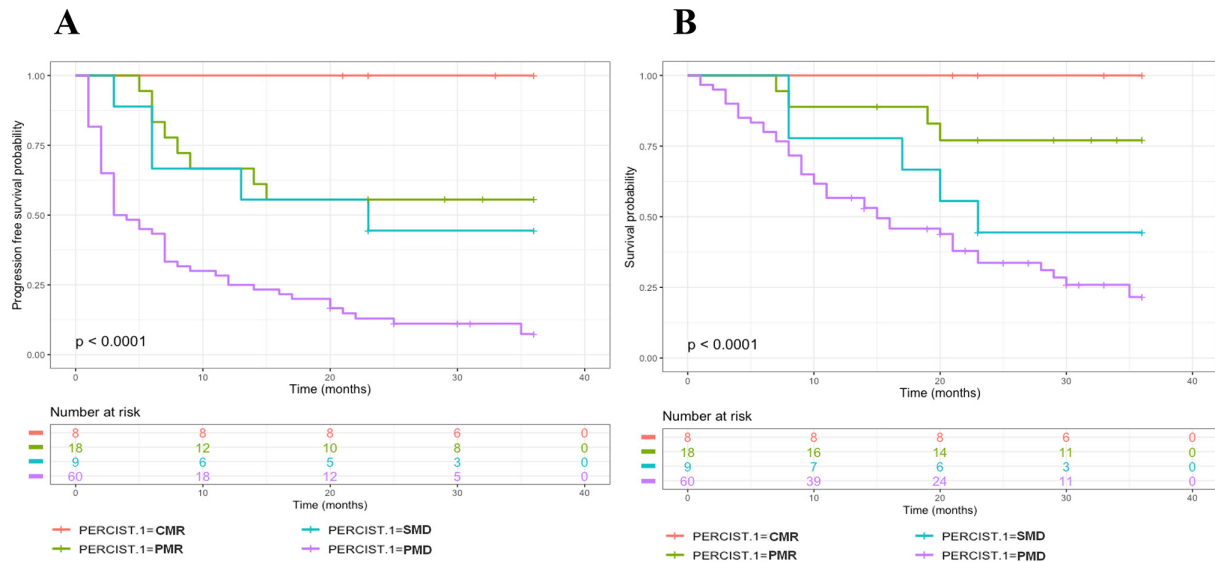


Figure 4 Progression-free survival (A) and overall survival (B) according to PERCIST evaluation at PET_{interim} 1. CMR, complete metabolic response; PET, positron-emission tomography; PERCIST, PET Response Criteria in Solid Tumors; PMR, progressive metabolic disease; PMR partial metabolic response; SMD stable metabolic disease.

- In patients with a PERCIST progression on PET_{interim} 1, the patients with a low TMTV₁ ($\leq 57.45 \text{ cm}^3$) (figure 6) had significantly better outcomes compared with patients with a high TMTV₁: 6 months PFS=73.1% versus 20.6%, respectively; 24 months OS=54.8% versus 17.9%, respectively ($p < 0.001$, both).

DISCUSSION

To our knowledge, this is the largest prospective study evaluating the prognostic value of metabolic parameters on both pretreatment and post-treatment ¹⁸F-FDG PET-CT

studies in patients with metastatic NSCLC receiving ICPI in monotherapy. It demonstrates the strong values of tumor volume-derived metabolic metrics assessed at baseline, as previously demonstrated,^{10,11} but also during early follow-up of patients after the start of ICPI.

Comparison with previous studies

Previous clinical and preclinical studies have highlighted the negative implications of a large tumor burden for cancer immunity.¹⁴ The total amount of cancer within the body can be estimated directly or indirectly by

PMD

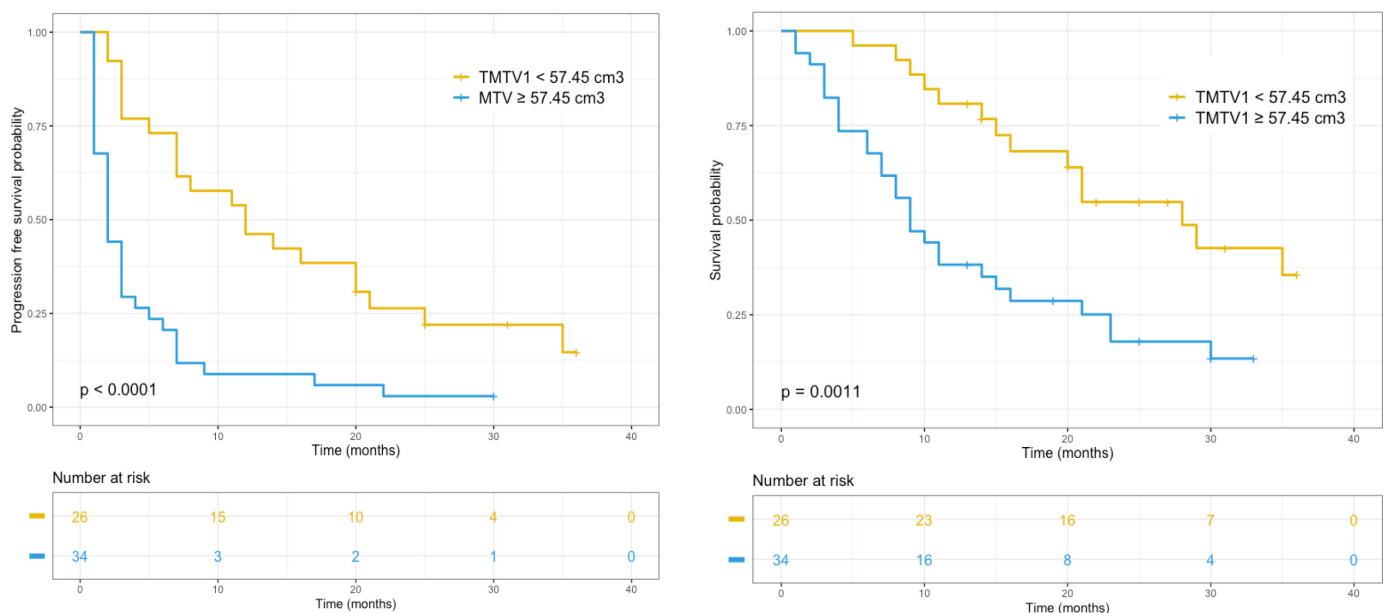


Figure 5 Progression-free survival and overall survival in PERCIST PMD subgroup on PET_{interim} 1 according to TMTV₁. PET, positron-emission tomography; PERCIST, PET Response Criteria in Solid Tumors; PMD, progressive metabolic disease; TMTV, total metabolic tumor volume.

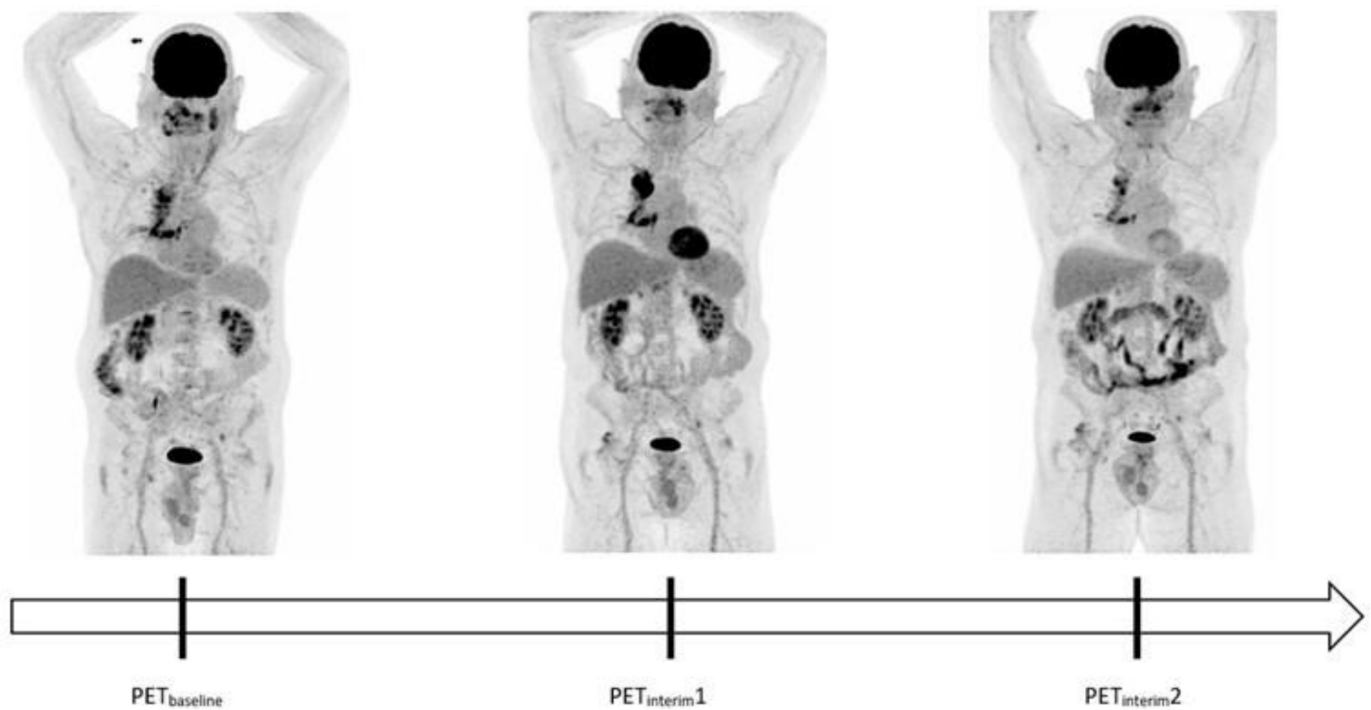


Figure 6 MIP images showing the tumorous evolution of a patient having a progressive disease according to PET Response Criteria in Solid Tumors (right pulmonary tumor) but a low residual total metabolic tumor volume₁ (<57 cm³) at PET_{interim1}. PET_{interim2} image shows a delayed decrease in tumorous burden. These low volume progressing patients could benefit from treatment continuation. MIP, maximum intensity projection; PET, positron-emission tomography.

various means, such as CT scans, ¹⁸F-FDG PET/CT, liquid biopsy methods (circulating tumor DNA or circulating tumor cells) or through the quantification of biological tumor derivatives such as lactate dehydrogenase (LDH). Previous studies demonstrated the adverse effects of a high tumor burden on the efficacy of ICPI that may be due to a higher glucose competition between tumor cells and CD8+T lymphocytes or a hypoxic tumor microenvironment, leading to a lower level of tumor-infiltrating lymphocytes in large tumors, CD8+T cell exhaustion and senescent immune profiles, as well as regulatory T cells impairment.^{22–25}

PET-CT is an effective imaging modality for accurate assessment of whole-body tumor burden as it usually covers the area from vertex to mid-thigh, thereby enabling all metastases to be imaged in a single study. The high tumor-to-background metabolic activity ratio also allows accurate identification and delineation of all tumor lesions, avoiding the limitation of non-measurable lesions.

In the present study, baseline MTV is a prognostic biomarker for 12-month OS but not for 6-month PFS. This better correlation observed on OS versus PFS is consistent with several other studies in the ICPI clinical setting.^{9–12, 26} It might be due to the uncertainty of determining true progression under ICPI, because of the well-known atypical evolutive pattern of PsPD,²⁷ leading to bias in the clinical definition of tumor progression to ICPI. To mitigate this issue, response status was systematically reviewed by a multidisciplinary review board, considering both clinical

and imaging aspects of cancer progression. The clinical aspects considered were mainly the ECOG score, signs of impaired general condition (asthenia, anorexia and weight loss), physical examination, pain assessment, and neurological evaluation. Routine laboratory values such as blood count, liver and renal function tests were considered. However, LDH was not systematically measured.

The optimal MTV threshold at baseline to distinguish between poor and good-prognosis patients differed between studies. In a retrospective study of 80 patients, Seban *et al* showed that baseline MTV was significantly associated with OS, using 75 cm³ as the best threshold, whereas the threshold was 5, 95 and 37 cm³ in the studies of Hashimoto *et al*, Monaco *et al* and Chardin *et al*, respectively.^{10–12, 26} These differences are due to variations between populations, PET/CT acquisition protocol and segmentation methods. For example, we applied the fixed absolute threshold of SUV>4.0, whereas other studies applied a fixed relative threshold of 41–42%^{10–12} or a threshold based on liver uptake.²⁶ We considered absolute fixed threshold as a better method for extracting clinically-relevant MTV information, such as its prognostic performance on consecutive studies, than relative fixed thresholds.²⁸ Indeed, using the fixed relative threshold method, a tumor with decreasing SUVmax between baseline and post-therapeutic studies may appear to grow in volume when in fact its boundaries remain the same, which is not suitable for treatment response assessment. It also may underestimate tumor lesions with heterogeneous uptake (eg, necrotic cores), which is often

observed in lung cancer. Another method, background thresholding, is more time-consuming as background uptake needs to be measured separately, lacks reproducibility and ends up being consistently around SUV 3–4 with liver-based thresholds, or around SUV 2 with mediastinal blood pool-based thresholds. Besides, in a prospective study by Tibdewal *et al* including 37 patients with operable NSCLC, MTV delineated with an absolute threshold of SUV of 4.0 correlated best with postoperative pathological tumor size.²⁹ More recently, the fixed absolute threshold of SUV>4.0 has also been designated as the method of choice in a prospective study by Driessen *et al* evaluating MTV delineation on lymphoma, requiring the least manual adaptation, allowing a high reproducibility and being observer independent.³⁰ According to the authors, this threshold should now be used in further studies to reach a consensus standard method. We also advise overcoming methodological inconsistencies between centers in the assessment of MTV by adopting this method as a standard for further prognostic studies in NSCLC.

Optimal predictive PET metrics and timing for the monitoring of response to ICPI

On the early post-therapeutic evaluation ($PET_{interim1}$), all metabolic parameters (SUV_{max1} , $TMTV_1$, TLG_1 , ΔSUV_1 , $\Delta TMTV_1$, ΔTLG_1) were prognostic biomarkers for both OS and PFS. In a previous prospective study on a more limited number of patients with NSCLC ($n=35$), $\Delta TMTV_1$ assessed at 8 weeks after ICPI initiation predicted OS only, whereas $TMTV_1$ and ΔSUV_1 predicted PFS only, with a quite similar optimal $TMTV_1$ threshold (84 cm^3 vs 57 cm^3 in our study).³¹

On the late post-therapeutic evaluation ($PET_{interim2}$), all metabolic parameters were still prognostic biomarkers for OS and PFS. This is the first study demonstrating such associations at this later stage. Considering the prediction for 6 months PFS, most $PET_{interim1}$ and $PET_{interim2}$ parameters had similar AUC values (range: 0.81–0.86). But considering the prediction of 12-month OS, $PET_{interim1}$ parameters showed higher AUC values than $PET_{interim2}$ ones (range: 0.81–0.83) (table 2). Because the metabolic parameters of $PET_{interim2}$ did not outperform those of $PET_{interim1}$, the early prognostic evaluation at 6–8 weeks after the start of ICPI appears to be more relevant than the delayed evaluation at 3 months. The clinical benefit of an even earlier PET assessment, for example, after a single cycle of ICPI, would be interesting to evaluate in future trials.

Concerning the selection of the best metabolic parameters on $PET_{interim1}$, volume-derived metabolic parameters had better performances than the standard ΔSUV_1 used in PERCIST criteria. Although $TMTV_1$ and TLG_1 achieve similar predictive performances, $TMTV_1$ seems a more robust biomarker as its calculation does not require to add the SUV_{mean} value, which lacks reproducibility among different imaging sites, PET systems and acquisition protocols.³² $TMTV_1$ and $\Delta TMTV_1$ had similar

accuracies for the prediction of 6-month PFS, but $TMTV_1$ exceeded $\Delta TMTV_1$ for the prediction of 12-month OS. Furthermore, the measurement of baseline $TMTV_1$ is not needed to obtain $TMTV_1$ and is thus easier to obtain than $\Delta TMTV_1$. Therefore, $TMTV_1$ may be the most relevant metabolic parameter for use in clinical practice, using a threshold around $50\text{--}60\text{ cm}^3$ of tumor burden.

Potential clinical utility of volume-derived PET parameters for patient's follow-up

How these PET/CT-derived biomarkers could benefit clinical decision-making on an individual patient's level has to be established. First, we demonstrated that $TMTV_1$ and PERCIST criteria are two independent prognostic factors for PFS and $TMTV_1$ is an independent factor for OS. Second, our study found a better prognosis stratification of patients using the residual tumor volume at $PET_{interim1}$ than using the standard PERCIST criteria, indicating a clinical utility for the patient follow-up after ICPI initiation.

Further subgroup analyses indicate that $TMTV_1$ remains a good prognostic biomarker for both PERCIST-responding and PERCIST-non-responding patients. In the non-responding group, a low $TMTV_1$ identifies a subgroup of patients keeping a very favorable long-term outcome (6 months PFS and 24 months OS rate=73% and 55%, respectively). A special attention is needed for these patients who can be identified as “progressing low volume” patients. Also, as PsPD cannot be distinguished on the first PET evaluating tumor response ($PET_{interim1}$), it may be relevant to consider the residual MTV in the decision of treatment continuation over a first PERCIST progression. It could avoid an early treatment discontinuation and a loss of chance in patients still having a clinical benefit of ICPI. Finally, for the few patients experiencing an early complete metabolic response ($TMTV_1=0\text{ cm}^3$), studies are needed to assess the interest of a potential therapeutic decay (dose reduction or spacing) as they have an excellent prognostic (24 months OS and PFS rate=100%, both).

Finally, we believe that to reach real clinical utility, $TMTV_1$ might be part of response prediction models combining several other factors that proved a predictive performance such as ECOG score, to improve therapeutic management.

Study limitations

Our study has limitations. First, the present study only included patients treated with ICPIs in monotherapy, while many patients with metastatic NSCLC are now treated with chemotherapy and ICPI combination treatment. Studies evaluating patients having this treatment regimen, which is a new standard of care, are needed to extrapolate our conclusions. While the median delay between a baseline PET scan and the initiation of ICPI was 7 days, nearly one in six patients had a baseline PET study more than 1-month prior to the start of treatment, despite the expectation that

PET findings would evolve over the course of a month for NSCLC. Furthermore, while this is the largest ^{18}F -FDG-PET prospective study on this clinical topic, it is monocentric, requiring a future multicentric validation. Moreover, the use of two different PET/CT systems may have affected quantitative SUV values, and therefore TMTV. The absolute fixed threshold used in our study might also overestimate MTV in tumor with very intense ^{18}F -FDG uptake, by spillover effect. Lastly, the manual segmentation of tumorous lesions, required for MTV measurement, is currently a limitation for its adoption in routine practice: it is time-consuming (around 15 min per study) and tedious. This difficulty should be easily overcome using artificial intelligence algorithms implemented in softwares, which are needed to get an automated and fast quantification of PET metrics such as MTV in NSCLC, as it has been previously used in studies on lymphoma or melanoma.^{33–36}

CONCLUSION

In patients with advanced NSCLC treated with ICPI, ^{18}F -FDG PET/CT volume-based metrics are prognostic non-invasive biomarkers. The absolute value of residual MTV, assessed after 6–8 weeks of treatment, is the optimal prognostic measure, and has a potential clinical utility. Oncologists should consider it in patients with a first tumor progression based on PERCIST criteria, as it identifies patients who could benefit from treatment continuation.

Correction notice This article has been corrected since it was first published online. An update has been made to the data in table 1 and table 2 and the headings of table 2. Reference 19 has been added to the first sentence of the *image analysis* section, which has re-ordered the reference list throughout the rest of the text. In the *patients' characteristics* section, 15 (instead of 12) of the 110 included patients did not undergo the PETinterim1 study because of early discontinuation of ICPI.

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Contributors All authors read and approved the final manuscript. PT is the guarantor of the study, collected the clinical and imaging data (especially PET metrics) and was a major contributor in writing the manuscript. DC performed the statistical analysis and contributed in writing the manuscript. NM collected clinical data. SC oversaw the data collection. FH collected imaging data. JO collected clinical data. OH conceived and designed the study and was a major contributor in writing the manuscript.

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Competing interests None declared.

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